

Predicting LendingClub Loan Interest Rates

Estimating Loan Pricing from Borrower Application Data

OPAN 6604 SAXA 5

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Note: We acknowledge the use of Anthropic Claude 4.8 opus, OpenAI GPT 5.5, and Google Gemini 3.5 flash in this project.



Context & Problem

Estimate a loan's interest rate at the moment of application, using only the borrower's financial and credit profile.

PROBLEM STATEMENT

- Predict loan APR for 10,000 held-out applications while minimizing prediction error (RMSE).
- Interest rates range from 6.6%-31%, with a mean of 12.95% and moderate right skew (0.85).
- Regression problem using structured application data only, with no time-series or text-based features.

ASSUMPTIONS & GUARDRAILS

- Loan status is a post-origination variable and was removed to prevent target leakage.
- Grade, sub-grade, and installment are not included in this dataset, requiring the model to infer pricing from underlying borrower characteristics.
- No temporal features are available, so model performance is evaluated using a random 80/20 train-validation split rather than time-based validation.

DATASET AT A GLANCE

100,000

Training loans

10,000

Test loans

38

Model features

6.46–31%

Target range

12.95%

Mean rate

Turning Applications Into Useful Signals

We translate each raw loan application into 36 consistent, comparable measures. These were applied the same way to every applicant and never using the answer we are trying to predict.

01 · DATA PREP

Fix data quirks; remove unfair hints

Credit strength - 6 features

Credit score, score tier, and loan length

A cleaner read on each borrower

Combined effects - 6 features

How score, debt, and loan length work together

Catches risk visible only in combination

02 · FEATURE ENGINEERING

36 clear, comparable indicators

Can they afford it? - 5 features

Loan size and debt compared to income

Weights borrowing against earnings

Credit history gaps - 7 features

Past issues, credit-file-age, missing records

Treats what is missing as meaningful

03 · VALIDATION

Checked on applicants it never saw

Risk groupings - 8 features

Debt and credit-use-tiers; extreme values smoothed

Sorts borrowers; tames outliers

Location and Purpose - 4 features

Home state, Zip Region, and reason of the loan

Adds local patterns without hints

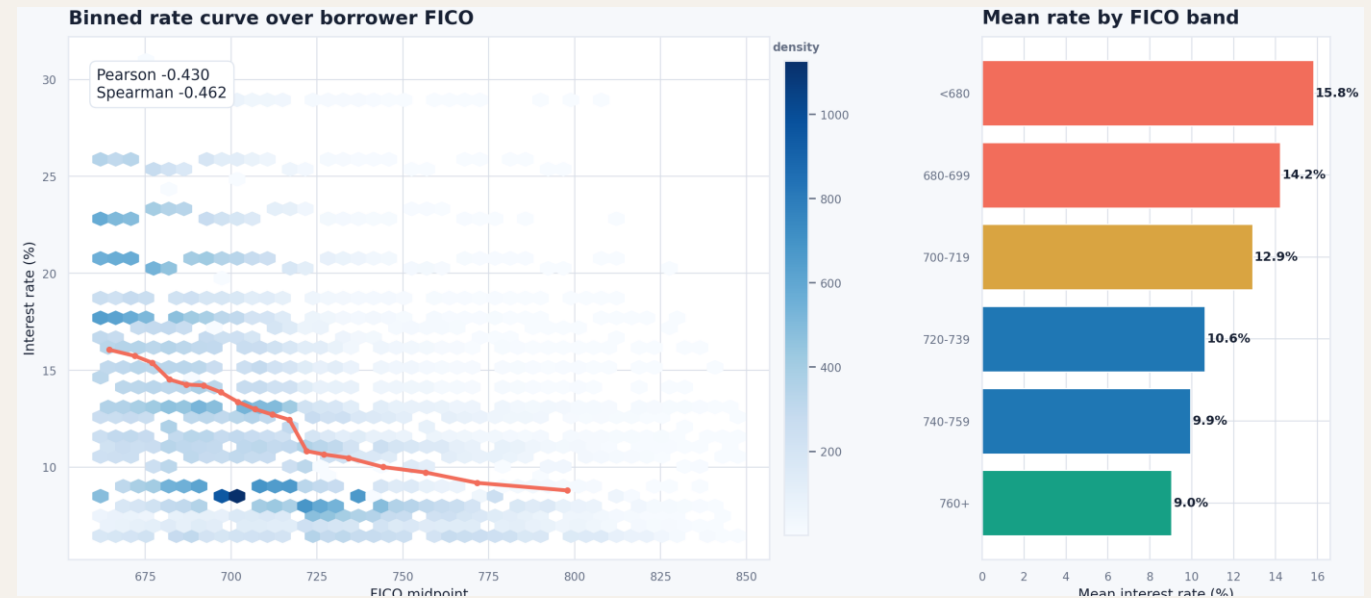
Guardrail: Loan Status dropped before modeling; encodings fit inside folds; raw FICO bounds, Term, Employment Length, Title & Zip Code replaced by engineered forms.

What the Data Says

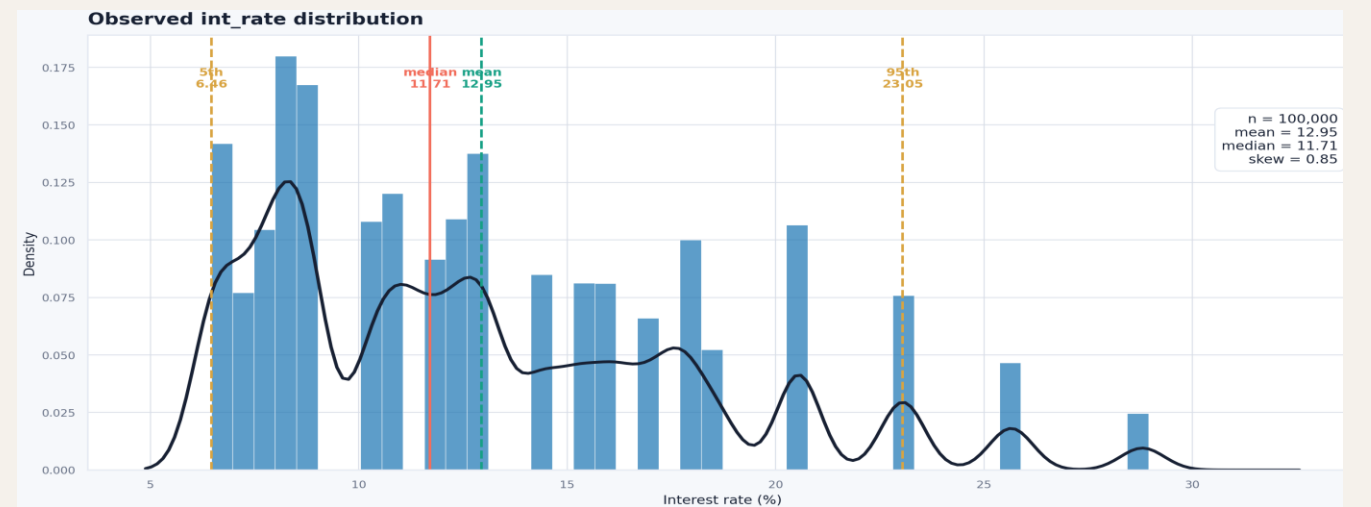
KEY FINDINGS

- FICO score is the strongest predictor of rate (Spearman -0.46), with utilization, DTI, and loan term moderately pushing rates higher (0.33, 0.30, 0.18).
- Longer 60-month loans and small-business or educational purposes tend to carry higher rates.
- "Months since last record" is missing 90.9% of the time, which itself appears to carry predictive signal rather than being just absent data.
- The test set is mildly different from training, slightly higher FICO scores and smaller loan amounts (PSI = 0.17), but not enough to undermine the model.

FICO vs INTEREST RATE

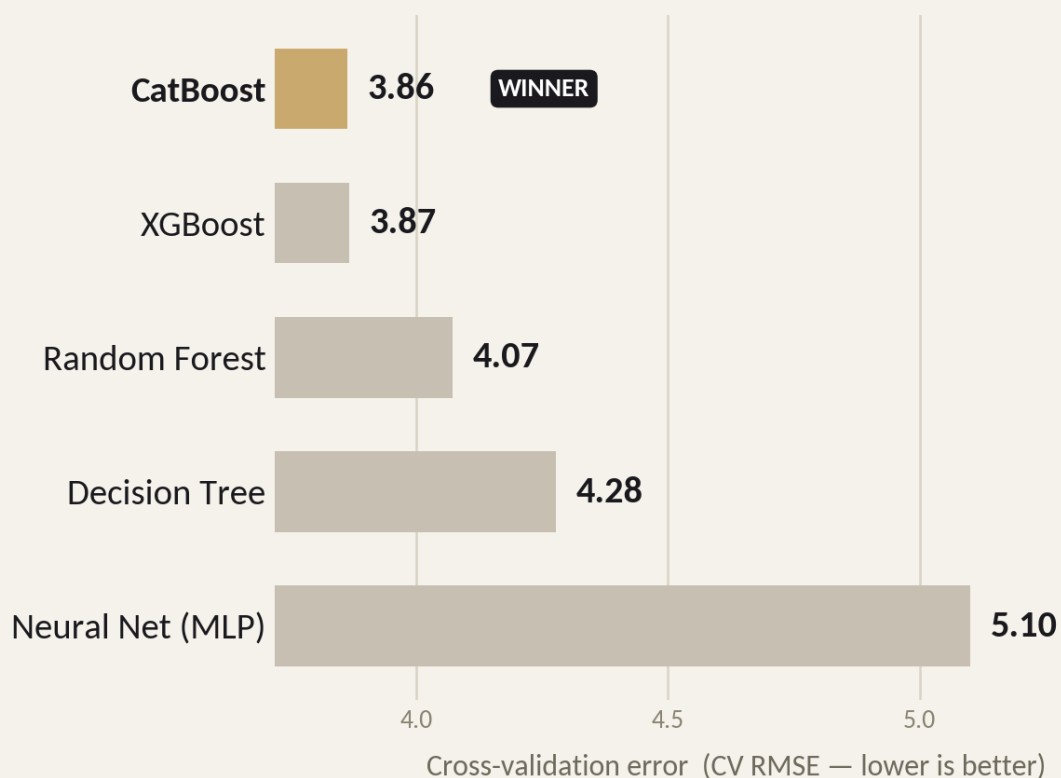


TARGET DISTRIBUTION



Five Models, One Winner

All five model families were trained on the same data and scored on the same hold-out set. CatBoost made the most accurate predictions.



THE WINNER

CatBoost

3.87

Prediction
error (RMSE)

2.90

Typical miss
(MAE)

0.465

Variance
explained (R^2)

- ✓ **8% more accurate** than a plain linear baseline (4.21 → 3.87).
- ✓ **Consistent, not lucky.** Its cross-check score (3.86) matches the hold-out (3.87).
- ✓ **Reads loan categories directly**, so less hand-tuning and fewer ways to leak the answer.

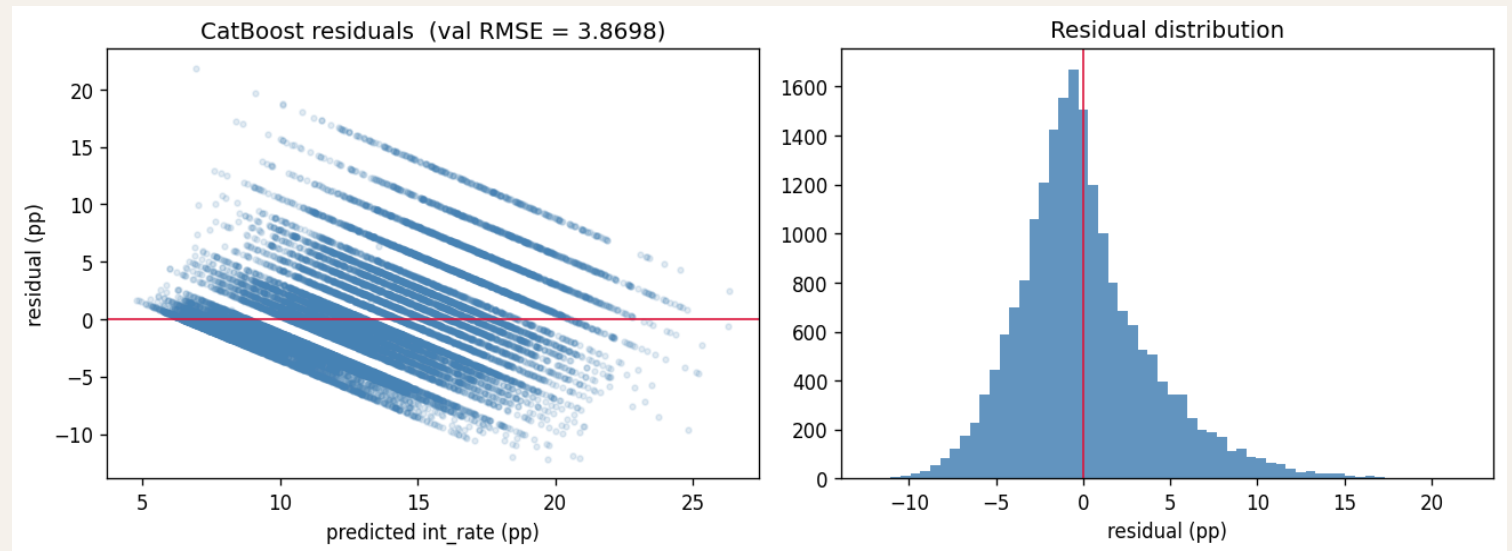
Final Model: CatBoost

CatBoost was selected as the final model because it handles categorical variables well and captures non-linear relationships in the data. Five-fold target encoding was used to reduce leakage and provide a reliable estimate of model performance.

TUNED HYPERPARAMETERS

Hyperparameter	Value
Learning rate	0.03
Tree depth	7
L2 regularization	5.0
Random strength	1.0
Bagging temperature	0.5
Border count	200
Max iterations	2,500 (early stop @ 100)
Predicted-rate range	6% – 31%

RESIDUAL DIAGNOSTICS



- Cross-validation RMSE (3.863) and validation RMSE (3.872) were nearly identical, indicating the model performs consistently on unseen data.
- That close agreement shows overfitting is not a major concern.
- The validation R^2 of 0.465 is reasonable given that grade and sub-grade, strong predictors of rate, were intentionally excluded.

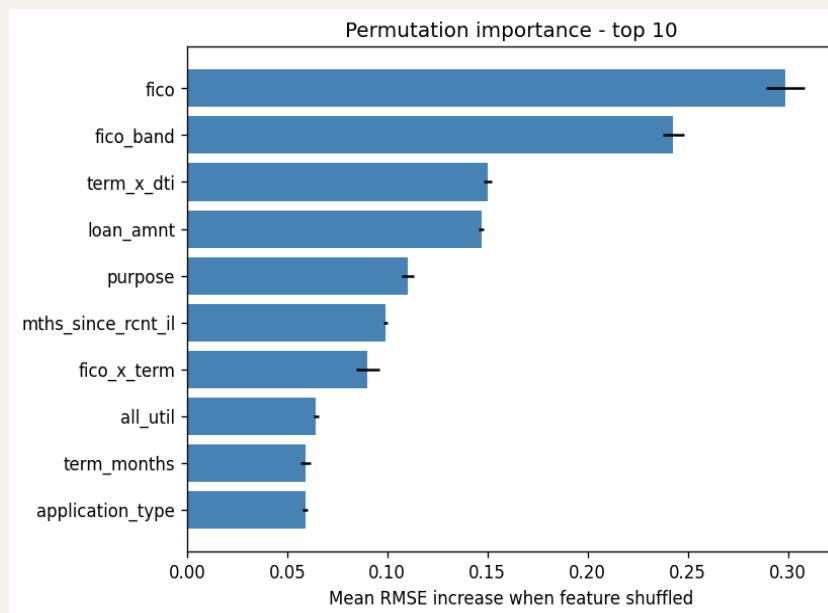
What Drives Interest Rate Predictions

The model's most important drivers align closely with the patterns identified during exploratory analysis. Credit quality, loan structure, and credit utilization have the greatest influence on interest rates.

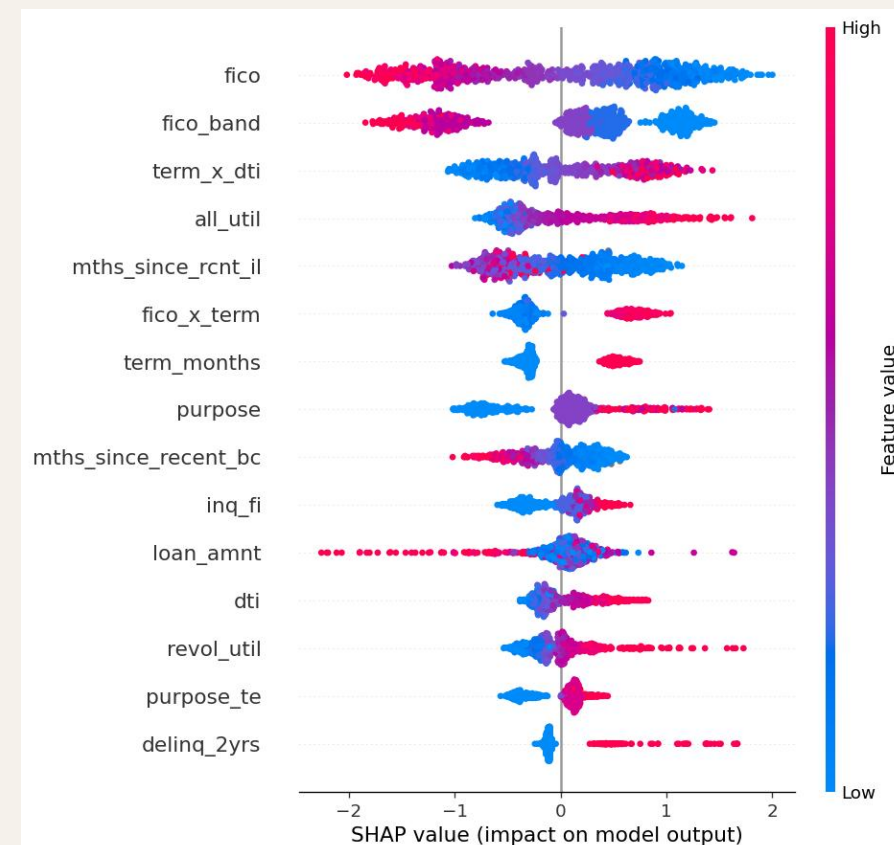
BUSINESS INSIGHTS

- FICO score was the strongest driver of interest rate.
- Loan term, debt-to-income ratio, and loan amount were the next most influential factors.
- Higher credit utilization and recent credit activity were associated with slightly higher rates.
- The consistency between SHAP values, permutation importance, and gain-based importance increases confidence in these findings.

PERMUTATION IMPORTANCE



SHAP SUMMARY



CONCLUSION · REFLECTION

Conclusion & Reflection

Application-time borrower characteristics provide enough information to estimate loan pricing with consistent accuracy.

± 3.9 pts

Typical error when estimating a loan's rate

47%

Of the differences in rates the model explains

Consistent

Same accuracy on new loans as in testing

WHAT THIS MEANS

- **Transparent by design.** It weighs the same signals a seasoned underwriter would, as credit strength, loan size, and borrowing habits are used so every estimate can be explained.
- **Instant, consistent pricing.** The rate a borrower qualifies for is estimated the moment they apply, so every applicant judged by the same standard.

WHAT WE LEARNED

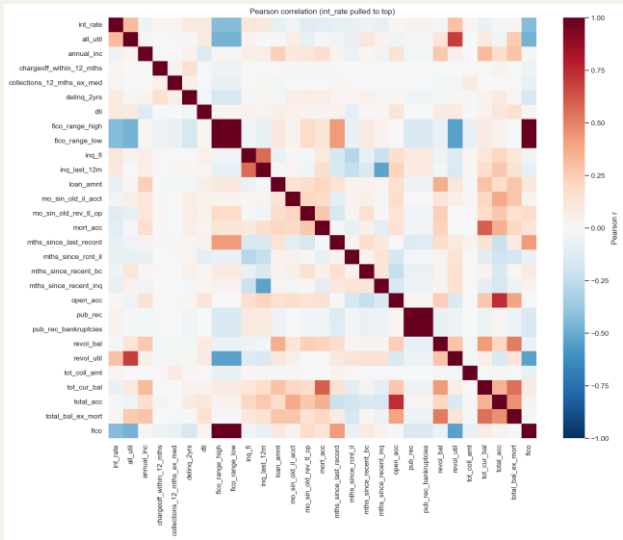
- **Clean data beat clever models:** Getting the inputs right delivered more value than any added complexity ever could.
- **Honest testing created confidence:** We measured the model only on data it had never seen, so the accuracy is real, not optimistic.

Final model: CatBoost, tuned for accuracy, with every prediction kept within the realistic 6%–31% rate range.

APPENDIX A

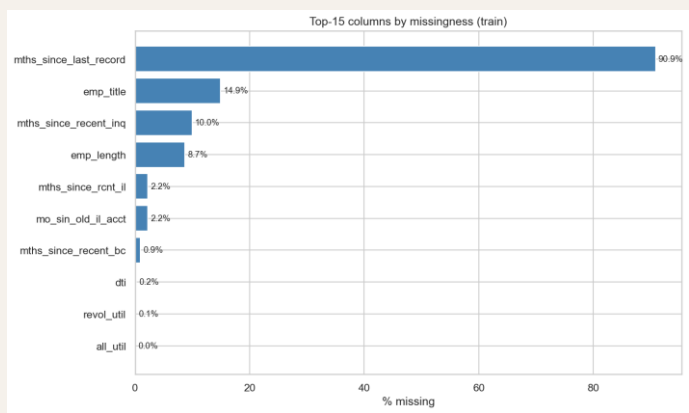
EDA Exhibits

CORRELATION STRUCTURE



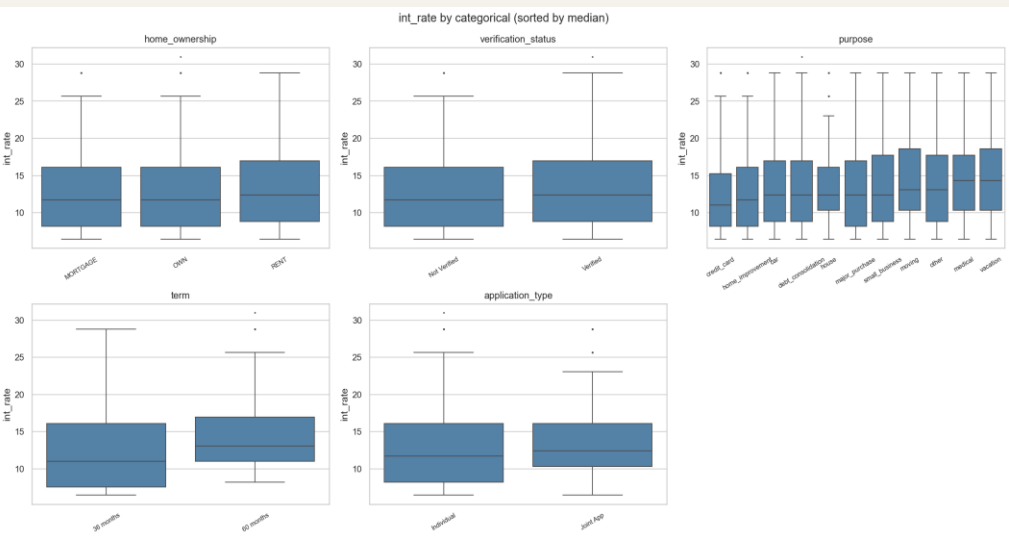
Most variables exhibit weak-to-moderate correlations, suggesting limited multicollinearity among predictors.

MISSINGNESS (TOP COLUMNS)



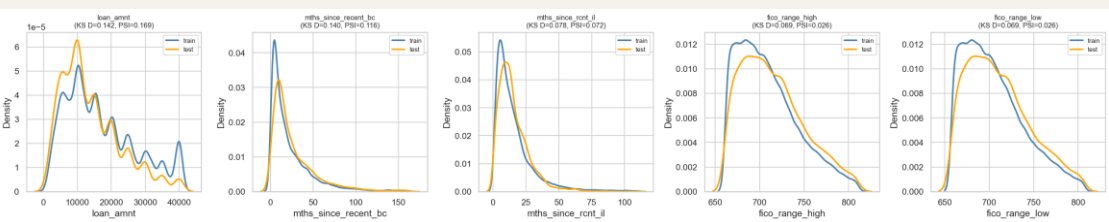
Missing data was concentrated in a small number of variables and addressed during preprocessing.

RATE BY CATEGORY



Interest rates vary across loan grades, purposes, and borrower characteristics, indicating meaningful differences between groups.

TRAIN vs TEST DRIFT

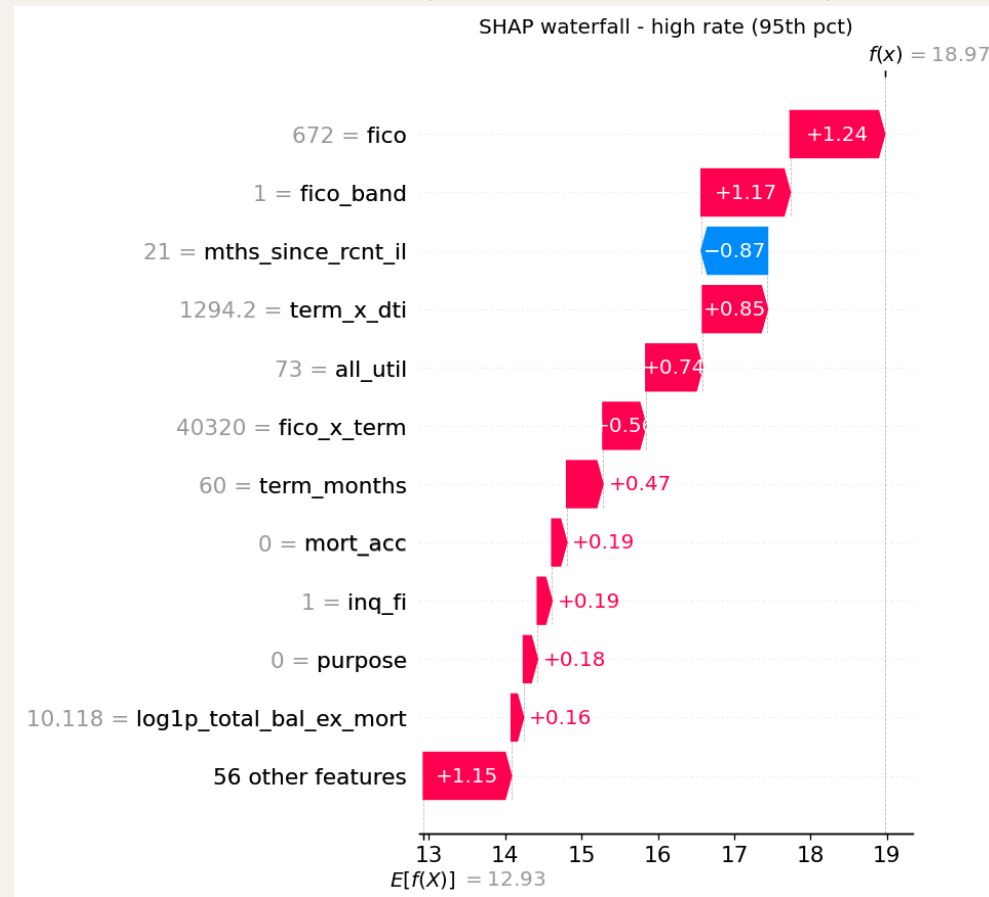


Key feature distributions remained similar between the training and test datasets, reducing concerns about dataset shift.

APPENDIX B

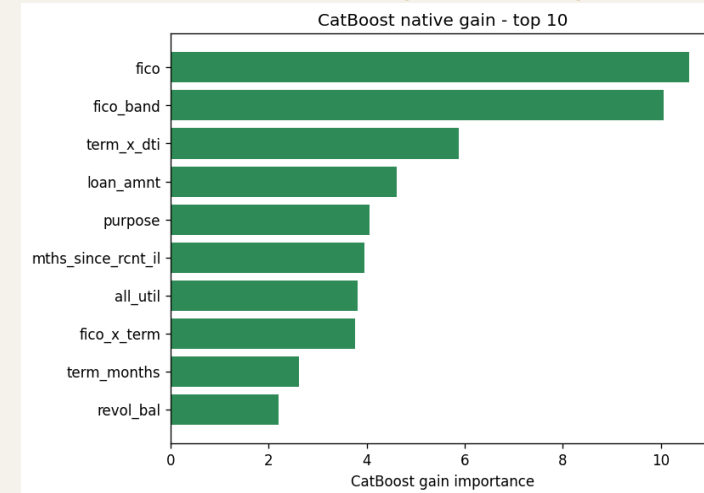
Explainability Exhibits

SHAP WATERFALL (HIGH-RATE LOAN)



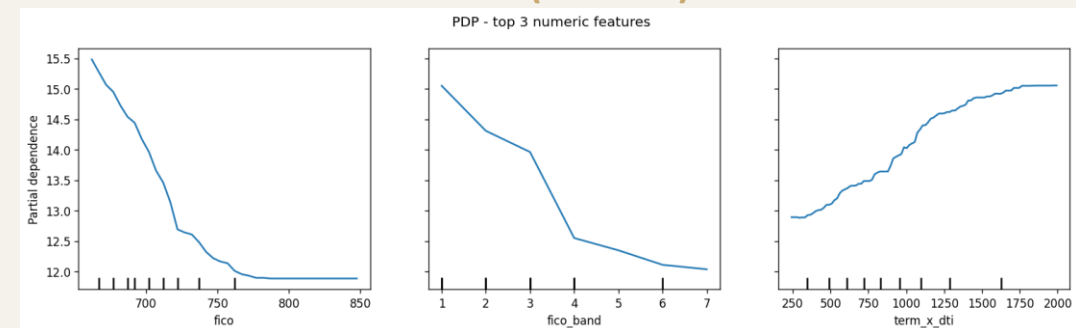
Low FICO and high term x DTI are the biggest upward pushers on this 95th percentile loan

CATBOOST GAIN (TOP 10)



FICO dominates tree-split gain; term x DTI and loan amount form a distant but clear second tier.

PARTIAL DEPENDENCE (TOP 3)



FICO's effect is steep and nonlinear, with the sharpest rate drops concentrating between 700-750.